

Guided Exploration Module 01: AI & Ethics

Points: 25

This is a substantial assignment, and Guided Explorations are worth **30% of your course grade**.

Grading: Guided Explorations are designed to demonstrate your understanding and application of course material. **Your responses should be grounded in the lectures and resources provided for the module and written in your own words.** Outside resources may supplement your learning but should not replace engagement with course materials. It is okay to make mistakes as you are learning. You will have future opportunities to apply these concepts, use feedback, and demonstrate your growth and improved understanding. **See the rubric in Canvas.**

Due Date: The final due date is listed in Canvas, but some parts may be due earlier as in-class participation. Follow the guides in Canvas and create calendar reminders to plan your time.

- Final submissions are accepted up to **24 hours late with a 10% deduction.**
- Submissions more than 24 hours late receive a **0.**

Submission: Upload your completed document as a **Word or PDF file** with your answers in the highlighted areas unless otherwise stated.

Objectives: See Module 1 Objectives

Collaboration and Academic Integrity

You are encouraged to collaborate by discussing concepts, asking questions, and exploring labs with others. **Your submitted responses should demonstrate your own understanding and be written in your own words.**

AI Use: AI Allowed with Disclosure

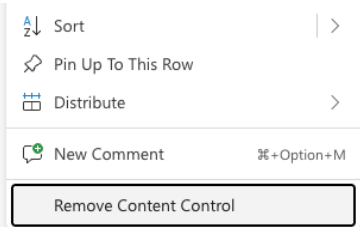
You may use generative AI as a **learning and research tool** for this Guided Exploration. AI may help you understand concepts, brainstorm ideas, locate resources, ask questions, troubleshoot, or get feedback on your thinking. **Your responses, analysis, and reflections must demonstrate your own understanding and be written in your own words.**

If you use generative AI, **disclose how you used it.** You are responsible for checking its accuracy, evaluating its suggestions, and being able to explain the work you submit. Do not copy and submit AI-generated responses as your own.

Use AI to support your thinking. Do not outsource the thinking the question is asking you to demonstrate.

Refer to the [CS Academic Integrity and AI Usage Policy - Harding](#)

If you download this as a word document and you have any problems typing your answer in the tables you must click on 3 dots and remove content control.



[Part 0 Intro: Learning and Collaborating](#)

[0.1 Working on Teams](#)

[0.2 Learning How to Learn](#)

[Part 1: Explore and Explain](#)

[1.1 Algorithms](#)

[1.1.1 Hello World](#)

[1.1.2 Inputs, Process and Outputs](#)

[1.2 AI Machine Learning](#)

[Part 2 Analyze and Apply](#)

[2.1 AI Systems](#)

[2.1.1 Bias](#)

[2.1.2 AI System Case Study](#)

[2.2 Teachable Machine Lab: How Do Machines Learn From Us?](#)

[Part 3 Reflection](#)

Part 0 Learning and Collaborating Reflection

The reality of computer science is that there are so many languages, technologies and methodologies available and it is constantly evolving with new ones. So the goal is not just to understand the current technologies but developing skills to learn any new technology.

0.1 Working on Teams

In this course, we'll work together on in-class activities, labs, and discussions. Good collaboration isn't just about dividing work — it's about building understanding together. Read [10 Collaboration Skills Examples](#) and answer the following

1 Pick two of the 10 skills that you think are strengths you bring to the team and how you will bring those strengths to the team.

- Teamwork because it is important to discuss with your team if there is problems or issues that need to be solved also, I like being in a team which we can work together and make things easier for everyone.

- Adaptability because I am open to hearing what my team has to say and trust them in leading also. I personally think being adaptive in many things can be a good thing because you will relate to the situation and know how to handle it.

2 Pick two of the 10 skills you think are most difficult for you and something you can do to try and improve in that area this semester.

- Giving and receiving feedback because I always haven't been good at that, but I am open to hearing what my team has to say and will work together. I also think
- Communication is tough for me too because I never start conversations. I think this must change because my degree is important to me and I also want to build more relationships with people not just the ones in my bubble.

0.2 Learning How to Learn

Learning how to learn is an important component of this course as your career will be full of learning something new.

1 Read [Bluesky CEO Jay Graber warns: If you're a student, using AI means](#) where Bluesky CEO's Jay Graber warns that over-reliance on AI can impair critical thinking and long-term skill development and may contribute to "academic obsolescence" - declining ability to perform without AI

What skills do you risk losing or not developing if you depend on AI too early? Give an example of how this could affect your learning.

I believe one risk is always relying on the AI to give your answer. I think this would affect my learning because if I were to just give AI all my work, I would never be learning anything, and it would just affect me in the future because I could get in the way of getting my degree and not learning anything.

2 Listen to this 30 minute podcast [Desirable Difficulties - The Learner Lab](#) or read [Student Resources - MSU Denver](#)

Which one did you choose? I chose the Student Resources from MSU Denver

Identify three strategies you will use this semester to strengthen your learning and use AI in ways that support—not replace—your own thinking. Briefly explain how you will use each strategy.

- Retrieval method could help me a lot this semester because I would have to use my brain to memorize things such as the stuff we learned about in class and put it into my work.
- Interleaving method is very important to me because when I study, I have the same problems, when mixing the types of problems from different

classes gives me a variety of options and I can always go back to the problem if I don't understand it the first time.

- Generating an answer before I see one. This is very important because when taking tests or quizzes I do the elimination process and it makes sense to me. But if there is a time when I don't know the answer, I just need to trust my gut and my instinct where I'm choosing the correct answer.

3. One of my favorite websites is [The Learner Lab](https://thelearnerlab.com). Explore The Learner Lab and choose one article, video, or podcast that interests you.

Include a link to what you chose.

<https://thelearnerlab.com/portfolio/do-hard-things/>

Describe one or two key takeaways and one specific way you could apply what you learned this semester.

One major takeaway from this video I learned was facing your fears and not just with spiders. This can be applied into class also. For me it's communication because I'm not always going to start the conversation because of what others might think but it's okay because others might not know it. It tells us about stepping out of our comfort zone and reaching for the things you want to reach and progressively getting better at it and overcoming your fears.

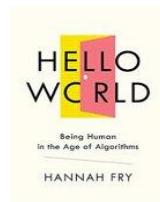
Part 1: Explore and Explain

Clearly explain key concepts from the lecture correctly in their own words containing examples from your work. Provides visuals (images, drawings, charts) and links to additional resources (videos, tutorials, etc.).

Review the [lectures](#) and links in the lectures to information. If you find you don't understand the information in the lecture reach out to **Deb or Alizah**

1.1 Algorithms

1.1.1 Hello World



"You can't just build an algorithm, put it on a shelf and decide whether you think it's good or bad completely in isolation. You have to think about how that algorithm actually integrates with the world that you're embedding in." Hannah Fry

Read [Chapter 1](#) from Hannah Fry, "Hello World, Being Human in the Age of Algorithms" and answer the following.

Answer the following in your own words.

1. Describe an algorithm. Algorithms is a step-by-step process which occurs and can lead to errors which also have solutions to them which lead to the output. For example, like when babies walk. The step is to crawl first, when they try walking up, they end up falling which are the errors and when they are old enough, they walk which is the output. But I also like to think of algorithms as repeated patterns or cycle of things which always continue.

2. Summarize the four main categories of real world tasks that algorithms are doing.

- First: prioritize their info and making a list so they know what comes first and what is important,
- Second: classification which looked at by algorithm in which people watch certain shows or movies giving them related topics the person is interested in.
- Third: finding new links to new things that the person hasn't seen or might be interested in clicking.
- Fourth: isolating what's important and that is what the algorithm puts on the screen, it may desensitize some content based on what the person has been watching and gives them more stuff they are likely to watch.

3. Explain the two main approaches taken by algorithms. Include the pros and cons.

The two main approaches taken by algorithms are machine learning and rule based. I would have to say the pros would have to be with the rule based right now because it's an actual human doing it so it would be more trustworthy. I would have to say a con for machine learning while it makes everything easier because of all the tools in the apps they will still have to deal with encountering errors and which would be more difficult for the machine to learn it which would take time.

4. What can happen when we rely on machines to make decisions for us?

When we rely on the machine to make the decisions for us, I think it puts the person more at risk on what they are sharing about themselves. Because once you put something on the internet it won't go away so letting machines do our thing is a serious risk of reputation for the person. Also, machines like to do it the first way they are told to with humans, they go through different trials and such which could take them longer to get a better result.

1.1.2 Inputs, Process and Outputs

Pick one of the following to describe the inputs, process, and outputs in your own words.

- A self-checkout machine
- A navigation app
- A weather app

What you picked from the three choices above: A weather app

Link to lecture that provides information on Algorithms.

<https://docs.google.com/presentation/d/1ZP5jvn84eJBq60YGFbSP9B-A3IKw6-a15PYRS8FeWA0/edit?slide=id.p#slide=id.p>

Inputs: for the inputs of the weather app, the first step is getting the location for the user. So, they need to start with the name of the city or can even be tracked using a GPS tracker to find where they are located.

Process (Algorithm):

Now with the process when the location has been found it generates an interface where when you load the app it shows smaller numbers and icons that people would tell instantly on what type of weather it would be. They would be able to see a pattern in the weather types as it changes from being sunny to rainy.

Outputs:

Lastly, for the outputs it goes specific to knowing how much precipitation the area can expect what's the chance of raining, and the temperature for each of the days listed on the weather app. It has a scaling from 0-100 on how hot or cold the day is going to be.

1.2 AI Machine Learning

Describe key ideas in your own words how machines learn and include examples. Use the lecture and resources in the lecture but put it in your own words.

1. Artificial Intelligence:

AI is a program that humans make and run it in a computer to do specialized tasks as humans would do but leave it up to the machine. This makes it seem like the computer has the humans mind and brain which makes it intelligent. One idea I saw is AI videos on my feed where there is fruits talking and AI was used to make that.

2. Machine Learning (ML)

Machine Learning is a set piece of AI that lets the computer run trials and errors so it can improve and perform better in the long term without it having to be programmed by humans. Which it reads patterns made by humans to see what they like and gives the content or feedback related to the persons interest.

3. Dataset

Dataset is a collection of information gathered in a folder or group which can hold anything in it. In the data numbers could be saved, images, videos, etc.

4. Learning Algorithm

To learn algorithms, it is just a procedure of steps that is needed to complete the task. Algorithms happen in the process part of the Input and Output. For example, baking a cake has its steps with how to cook it.

5. Prediction

A prediction is like a hypothesis. It reads what is about to happen or could even happen in the future without being certain about it. The machine tries to guess and predict the result with the information given to it already.

6. Large Language Models (LLMs): LLM's to my knowledge are even more advanced AI system that have a larger scale to what they are trained on. For example, using large amounts of texts and things like that while using messages they can predict what the next word you might type or might even spell it out for you.

7. Neural Networks

A neural network is the learning model of how it recognizes patterns and such. Like how we were talking about algorithms, it can input content based on what you have watched before. And keep that thing going.

8. How do AI-driven systems differ from simple automated systems? Give an example of each.

I think the difference between AI and automated system is that AI is more of a problem-solving tool for example when trying to program something it gives you a syntax error and tries to fix it right away. While as automation it already has a script it must follow, and it just repeats the steps until it is stopped or has broken down.

9. Resources used such as link to lectures and websites

https://docs.google.com/presentation/d/1ZP5jvn84eJBg60YGFbSP9B-A3IKw6-a15PYRS8FeWA0/edit?slide=id.g37188fc7d0f_0_79#slide=id.g37188fc7d0f_0_79
<https://drive.google.com/file/d/1UNweLQrXpyFJ41N1lYy2ypCPp-9F41uu/view>

Part 2 Analyze and Apply

Demonstrates a clear analysis of a problem and applies relevant lecture concepts effectively. Work shows independent thinking and ethical use of AI tools or external resources.

2.1 AI Systems

2.1.1 Bias

Watch [Algorithmic Bias and Fairness #18 | PBS](#)

Compare how AI systems across different fields demonstrate bias. Read the articles from the resources, then in your own words complete the chart below by identifying:

- Bias Identified: What kind of unfairness, error, or imbalance occurred?

- What was affected: Be specific: who experienced harm or exclusion?

System and Type of AI	Bias Identified and Who Was Affected
Facial Recognition AI Face detection	NIST NIST Study Evaluates Effects of Race, Age, Sex on Face Recognition Software 2019 Arxiv Review of Demographic Bias in Face Recognition2025 Bias: Higher error rates on darker-skinned faces Affected: Black women and nonwhite subjects
TikTok Feed AI Content Recommender	University of Minnesota How is TikTok affecting our mental health?2023 UCL Social media algorithms amplify misogynistic content to teens (2024) Optional Video - How TikTok's Algorithm Figures You Out WSJ Bias: The content on TikTok is being fed to users which has a impact on their mental health and lifestyle. Affected: Users on TikTok especially teenagers. Harming them because of the daily scrolling users do and the content that is being given to them continuously.
Hiring Tool AI Resume screening model	Reuters Amazon scraps secret AI recruiting tool that showed bias against women 2025 University Washington 2024 Bias: Amazon AI hiring tool that doesn't want to give job opportunities to some people in need of jobs. Affected: Women looking to get jobs at Amazon. They were excluded because a machine didn't select them.
YouTube Recommendations AI: Content ranking model	Mozilla YouTube Algorithm Recommends Videos that Violate the Platform's Very Own Policies 2021-2024, Brookings Echo chambers, rabbit holes, and ideological bias: How YouTube recommends content to real users 2022 Bias: Content on YouTube is disturbing and hateful which influences violence and is a rule against their own policy. Affected: Users on YouTube and mostly non English speakers in other countries. It was harming them because of

the content they were seeing. It can mess with their brain and make them also vulnerable.

1. How do biased training data or algorithms create real-world harm?

They create algorithms because they only go off of what they know and the aspect of the profile. Like for example the Amazon hiring was only choosing men to work probably because of the structure they put the worker as.

2. What role should humans play in checking these issues?

I think humans should play the role after the AI checks it because if they are a double check person it makes it easier for the Ai and the human because they are both playing a crucial role and makes things easier.

2.1.2 AI System Case Study

Review [Self Driving Cars Case Study](#)

In this activity, you will choose a real AI system and explore how it works. You will research and explain how your AI system performs each of the five parts of intelligence: Perceive, Reason/Plan, Learn, Interact, and Impact. Then analyze the system from different perspectives

Choose one example from above or find your own example to explore how the AI system works. Here are some other examples

- ChatGPT or another chatbot
- Siri, Alexa, or Google Assistant
- AI used in medicine (AI diagnosing diseases)
- AI in gaming (AI opponents in video games)
- AI recommendation systems (Netflix, Spotify)

When searching, try typing questions like:

- "How does [AI system] work?"
- "How does [AI system] make decisions?"
- "What data does [AI system] use to learn?"
- "How does [AI system] affect people and society?"

5. AI System you chose to do a case study and brief description: AI in gaming

Include the resources you used below. If AI suggests a resource, open and evaluate the original source before using it.

6. Explain how it works for the AI big Ideas. You do not need to provide technical answers to the questions but include 2 to 3 sentences in your answers.

6.1 Perception: What does AI need to observe? What tools (text, cameras, microphones, sensors) do it use?

The AI needs to observe how the bots play out with the environment in the games. For example, making the character in the game sprint, hold a weapon, jump, etc.

6.2 Representation and Reasoning: What decisions must be made? How does it figure out what to do next?

They need to program the bot in the game to do this part. But for most games the bots aren't controlled by humans, so the functions are already generated in them so when they see an enemy they attack or interact.

6.3 Learning: How does it get better over time? Does it use data or feedback to improve?

To improve the bots in the game they need to give it more functions. Functions like running are basic and which they need. For the bot to do more in the game like I said they need more functions in their program.

6.4 Natural Interaction: How does it take action or communicate? Does it talk, move, show results?

Mostly in cutscenes they talk. When a person plays the game, it feels like a movie. Giving the functions to a bot can really change how the game plays out and how the person playing it experiences it also.

6.5 Impact: How does this AI affect people and society? What are the benefits and possible problems?

I feel like this would affect society because people have a chance to build emotion relation with this. And this is bad because this can lead to people having relationships within game characters.

7. Impact from different perspectives. Pick at least 3 of the following perspectives to answer in 2 to 3 sentences.

Scientific: Is it accurate and reliable? I would say this is reliable because I've personally play video games with AI and I see it in the games with the bots. This has been a part of video games for years now and it gets better with certain parts but could be bad with others too.

Economics: Who can afford it? Who profits? Anyone willing to buy a certain game that has AI can own it. It is very easy to get your hands on games with these features and can be downloaded digitally. The game company profits and the game carrier companies like PlayStation, Xbox, Steam.

Global: Can people access it worldwide? I would say it depends because there are certain games that aren't allowed in some countries. But depending on the game it

can be worldwide.

2.2 Teachable Machine Lab: How Do Machines Learn From Us?

Feel free to collaborate to do this lab with 1 to 3 other people.

Names of collaborators: Thomas

Use the <https://teachablemachine.withgoogle.com/> to create machine learning models. Choose one project type below: Image

- Image - [Teachable Machine Tutorial: Bananameter | by Barron Webster | Medium](#)
- Audio - [Teachable Machine Tutorial: Snap, Clap, Whistle | by Barron Webster | Medium](#)
- Pose - [Teachable Machine Tutorial: Head Tilt | by Barron Webster | Medium](#)

Train your model

1. What type of input are you training the model on (image, audio or pose)?
Image

2. List the categories you will teach the machine (Example: Clap vs. Whistle, Banana vs. Apple) and how many examples (recordings/photos/poses) added for each category

2.1 Category 1: mobile phone

2.2 Category 2: laptop

2.3 Why is it important to include different versions (voices, angles, lighting, etc.)?

It is important to include different versions because if you take your photos in the same place every time it might recognize something else instead of the laptop or phone which is the whole point of this project.

3. Model Training

3.1 What do you think the computer is learning from your examples?

The computer is identifying the differences between the laptop and phone. For example, the shapes, lighting, where it's placed, size, etc.

3.2 What would happen if you only added one example per category?

I think what would happen is it would just try to figure out what it's looking for., For example if we had a phone with a chair and just one example it would also recognize the chair too.

4. Testing:

4.1 Did the model correctly identify each input when you tested it? (Yes, most of the time No, it makes mistakes)

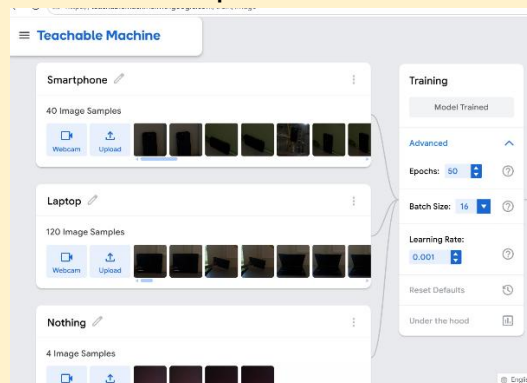
Yes, every single time we held the phone or laptop at different angles and lighting and still spiked up when we showed it.

4. Test with different datasets (A new image or sound, A different voice or pose) Did it still work? Why or why not?

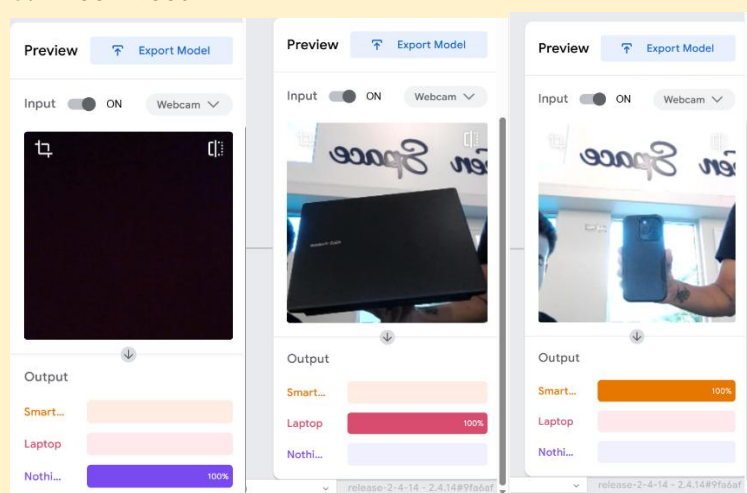
It didn't work with a pose because when I was holding the laptop it was confused on whether it was a laptop or a mobile device.

5. Take a screenshot of your model setup and one test result. Use "snipping tool" or screenshot shortcut; paste below then crop and resize to make it easy to identify.

5.1 Model Setup



5.2 Test Result



6. Improve Your Model

6.1 What's one thing you could do to make the model more accurate?

I think one more thing I could do to make the model more accurate is maybe put it in the same place where we took the original picture.

6.2 Do you think this model would work with other input, in different lighting or with different accents?

Yes, I still think it would work because it would still recognize the shape and size also the design of the phone and laptop.

6.3 What's one limitation of the model you noticed?

One limitation is what we had in the example photos, for example if we are standing in front of the camera while holding it, the machine gets confused and doesn't recognize it.

7. Learnings: Explain in your own words how machines learn using correct terms from lectures and include examples from this lab.

Machines learn by identifying patterns and recognizing similarities given from images, audio, or videos. For example, like the activity we did with the drawing it was recognizing patterns and trying to guess what it is from beginning to end.

Part 3 Reflection

Artificial intelligence is becoming part of everyday life — from hiring decisions and healthcare to video recommendations and facial recognition. But with that power come serious ethical concerns.

Responses demonstrate in-depth personal reflection on the experience and learnings that includes plans for future action.

1. In your own words, explain two ethical issues that can arise from using AI .For each issue, suggest at least one step that designers, companies, or governments should take to reduce harm or promote fairness.

One issue is people getting too much information fed to them from the media which is bad and another one is also how AI violates our privacy. A solution to this is not to give people, especially kids, all that information from media so they can filter content and cause less harm.

2. AI in your life

2.1 Give 1–2 examples of how AI shows up in the apps, tools, or services you use.

- Clothing brand ads including AI on social media
- AI automated messaging system when buying from companies such as Amazon

2.2 What did you learn in this module that might change the way you use or think about AI?

I learned that many people use AI to help them but also use it in a unethical manner which can lead to many bad things happening to them or society.

3. Responsibility and Accountability

Who do you think should be responsible when an AI system causes harm — the developers, the company, the government, the users? Explain your reasoning with an example.

I think whoever is using it should be responsible because it was their choice to use it unethically.

4. AI Use Disclosure

Did you use generative AI while completing this Guided Exploration? **Yes / No**

If yes, briefly describe how you used it and give one example of how it helped—or did not help—you're learning.

Yes, I did use it when doing the computer vs mobile phone experiment. I used AI to give me ideas on where to place the devices. It told me lighting mattered otherwise the machine learning tool wouldn't be able to tell.